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A diagnostic for measuring H_2O_2 concentration in a shock tube using tunable laser absorption near 7.8 μm

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Abstract

Hydrogen peroxide was measured in a shock tube via absorption spectra of the ν_6 (asymmetric HOO bending) fundamental band using a lead–salt diode laser tuned to 7845.63 nm. At this wavelength, there is minimal interference from water absorption at combustion temperatures and pressures, even for water levels that are more than two orders of magnitude larger than the H_2O_2 concentration. For obtaining a calibration of the laser absorption technique, a robust method of hydrogen peroxide delivery was employed so that a standard H_2O_2–H_2O mixture could be used as the source, but with high certainty as to the initial concentration of H_2O_2. The initial amount of hydrogen peroxide was measured using a water absorption laser diagnostic to measure the water concentration in conjunction with the known thermodynamics of the vapor mixture, highly diluted in argon. Measurements were made with argon as the bath gas at temperatures ranging from 292 to 1296 K and a pressure range of 0.05–1.43 atm, resulting in a calibration for the absorption coefficient of H_2O_2 at 7845.63 nm. To the authors' knowledge, this study marks the first successful measurement of H_2O_2 concentration in a shock tube using tunable laser absorption in the infrared.
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Keywords: Hydrogen peroxide; Laser absorption; Shock tube; Infrared; Chemical kinetics

1. Introduction

Hydrogen peroxide is formed in significant quantities during the combustion of hydrogen and hydrocarbons at intermediate temperatures (850–1200 K) and elevated pressures, and it plays a large part in the production of hydroxyl radicals prior to the onset of ignition. As mentioned in work by Westbrook [1] and others, in this moderate-temperature regime below about 1000 K an accumulation of H_2O_2 is produced, and its subsequent thermal decomposition causes ignition. This chain branching H_2O_2 decomposition, $H_2O_2 + M \rightarrow OH + OH + M$, becomes important at intermediate temperatures but is less so at higher temperatures (above 1200 K) as the reaction itself quickens, and the H_2O_2 is consumed as fast as it is

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formed. The accumulation of hydrogen peroxide and the importance of H_2O_2 decomposition is outlined in the study by Griffiths et al. [2] attributing its relevance to multiple-stage ignition and low-temperature chemistry in simulations of a rapid compression machine (RCM). Some interest in H_2O_2 also comes from the atmospheric sciences as it contributes to the formation of sulfuric acid in rain water, as stated by Bechara et al. [3] in their work with measuring hydrogen peroxide using a tunable diode laser (TDL).

To date, most of the experiments performed on hydrogen peroxide at elevated temperatures were completed in static cells at temperatures ranging between 600 and 900 K [4–7]. Also, hydrogen peroxide is notoriously difficult to work with. For higher temperatures, shock tubes have been used to investigate thermal decomposition of hydrogen peroxide at 1000–1200 K by only two groups as of this study, the laboratory of Troe and co-workers [8,9] and recently by Hong et al. [10] at Stanford University. The research by Kappel et al. [8] and Meyer et al. [9] was conducted using UV absorption spectroscopy of H_2O_2 at 215 nm with a frequency-quadrupled Ti:Al_2O_3 laser source, and at 215, 230, and 290 nm with a UV lamp. Considerable difficulties resulted in these UV regions with either the absorption of HO_2, which absorbs strongly at transitions around 215 nm, or reduced intensity from the UV light source. In these studies, a vaporization process on 70% wt. H_2O_2 solution yielding 99% pure hydrogen peroxide was used for measurement. The research of Hong et al. looked at the rate of the decomposition reaction by employing an extremely sensitive water absorption spectroscopy diagnostic at 2550.96 nm to measure the production of H_2O from the hydroxyl recombination reaction, from which the decomposition rate of H_2O_2 could be inferred. Hong et al. used urea–H_2O_2 as the hydrogen peroxide source.

Ideally, a direct measurement of H_2O_2 concentration in a shock tube using laser absorption would be preferred when studying the chemistry of hydrogen peroxide under combustion conditions where H_2O_2 plays an important role. However, two problems arise when developing and using such a H_2O_2 diagnostic: (1) obtaining a reliable source of hydrogen peroxide for calibrating the technique and (2) isolating an appropriate spectroscopic feature that is free from interference with HO_2 and H_2O. In the present paper, we present an approach that successfully resolves both of these problems.

Reviewing available data from the HITRAN 2004 database [11] at specific frequencies in the infrared, there exist only narrow regions where the H_2O molecule will not interfere with the absorption of hydrogen peroxide. Figure 1 highlights a specific spectral region of interest near 1275 cm^{-1}. In Fig. 1b, a line has been chosen for the remainder of this work at 1274.595 cm^{-1}, a transition which exists in the fundamental ν_6 band of the H_2O_2 molecule; this region was chosen because it coincides with a region that is accessible to a lead–salt diode laser, which is described in more detail in Section 2.2. Figure 2 shows the lineshape details for the feature of interest herein,

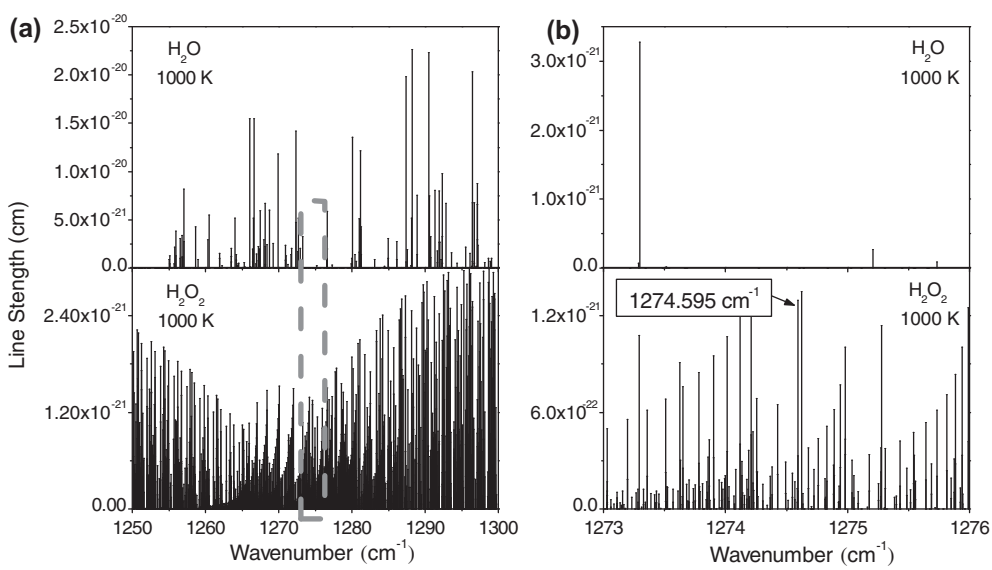

Fig. 1. Line strength data from HITRAN 2004 for water and hydrogen peroxide at 1000 K for (a) broad scale of wavenumber to show rovibrational patterns with a dashed box detailing part (b) which is a closer view outlining the chosen H_2O_2 line.

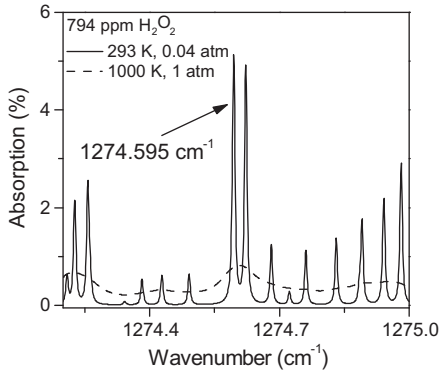

Fig. 2. Absorption of 794 ppm H_2O_2 with $X_{H_2O} = 0.0316$ and argon at conditions leading to 1 atm behind the shock wave.

calculated using a Voigt fit and the spectroscopic information in HITRAN. For Fig. 2, broadening between H_2O_2 and H_2O was estimated from the work by Devi et al. [12] and for H_2O_2–Ar by utilizing similar qualities to H_2O–Ar in the work of Nagali et al. [13] as they are not currently available in the HITRAN database. The temperatures and pressures represented in Fig. 2 were chosen to reflect those that are representative of the shock-tube experiments presented in the present paper.

Although HITRAN provides a good estimate for H_2O_2 spectroscopy in the region of interest, there still exists a lack of fundamental spectroscopic data for hydrogen peroxide at elevated temperatures and pressures in the infrared. It was the impetus of the present study to develop an accurate laser absorption technique for measuring H_2O_2 at combustion-relevant conditions. Details on the shock-tube technique are presented first, followed by a discussion on the lead–salt diode laser. An accurate method of generating a known amount of hydrogen peroxide in the shock tube is presented next, along with the measurements leading to an accurate calibration and plans for future work.

2. Experimental details

2.1. Shock tube

The shock-tube facility used in this work is described elsewhere [14] and is briefly detailed herein. H_2O_2 concentration measurements were made in a stainless steel 304 helium-driven shock tube with a driven section inner diameter of 16.2 cm and length of 10.7 m. A schematic of the shock-tube facility with details relevant to this study is presented in Fig. 3. Ultra-high-purity argon is passed through a saturator containing an aqueous solution of H_2O_2 and H_2O (either 50% or 35% by wt. H_2O_2) provided by Fisher Scientific, and the saturated mixture is sent through the shock tube in a process described in greater detail later. Pressure of the mixture prior to the shock is determined by a high-accuracy MKS Baratron pressure transducer. Post-shock conditions are determined using the standard one-dimensional shock relations with the measured incident shock-wave velocity. Uncertainty in the post-shock temperature in this facility is determined to be 10 K [14].

2.2. H_2O_2 laser diagnostics

A Laser Components (Laser Photonics) TDL system with L5830 TDL controller provides the infrared source beam to be absorbed by selected transitions of the ν_6 band of hydrogen peroxide in the 8-μm region of the spectrum. The lead–chalcogenide semiconductor laser source is mounted inside a dewar on a liquid nitrogen-cooled plate. Diode temperature is controlled using local heating current and a feedback loop. The diode requires injection current above 1000 mA to reach lasing threshold in the spectral region of interest, therefore the controller was upgraded with the ability to source up to 2000 mA. A third parameter set by the controller is the waveform and amplitude of injection current modulation. The laser, optimized for use at ~1320 cm^{-1}, is tunable in short segments of a fraction of a wavenumber, after which another laser cavity mode takes over. The typical length of a continuous scan is about 0.6 cm^{-1}, and the tuning rate is typically about 0.014 cm^{-1}/mA. When the mode switches, the new frequency is displaced by about 3.5 cm^{-1} – the separation between cavity modes – so that the majority of the spectrum is missed.

The complete shock-tube laser spectrometer is diagrammed in Fig. 4. The infrared beam exiting the diode laser is collimated by an off-axis, parabolic mirror. A small portion of the collimated beam is picked off with a barium fluoride window and further split into N_2O and germanium etalon legs for spectral frequency calibration. A monochromator internal to the system is bypassed in the present configuration. The infrared beam is focused near the shock-tube entrance window with a spherical mirror of 50-cm focal length.

A single Judson HgCdTe infrared detector with matched preamp was used to generate time-dependent absorption profiles. An SR560 differential preamplifier provided low-pass filtering with 300-kHz cutoff to eliminate high-frequency noise from the preamp, and further boosted the signal amplitude.

Elimination of other laser modes can be accomplished by passing the laser beam through a monochromator optimized for transmission of the principal mode and adjusted for narrow bandpass. The disadvantages of this approach are power loss that can be significantly greater than

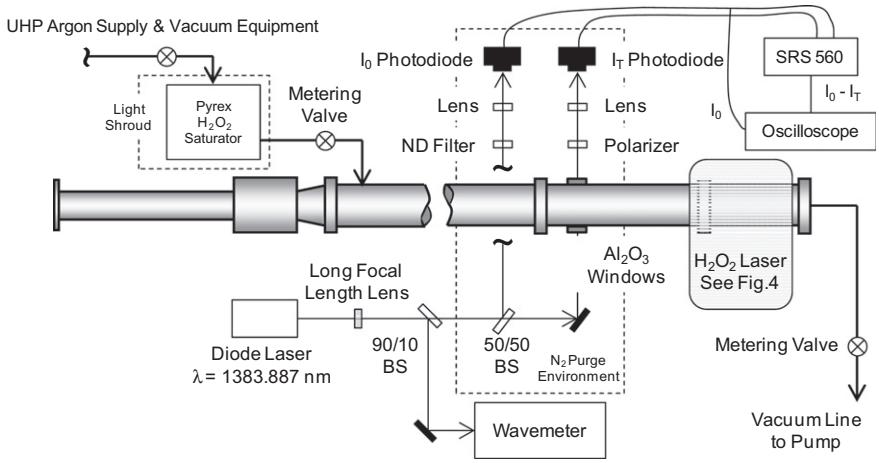

Fig. 3. Schematic showing H_2O_2 delivery system, water laser diagnostic, and relative position of H_2O_2 laser diagnostic detailed in Fig. 4.

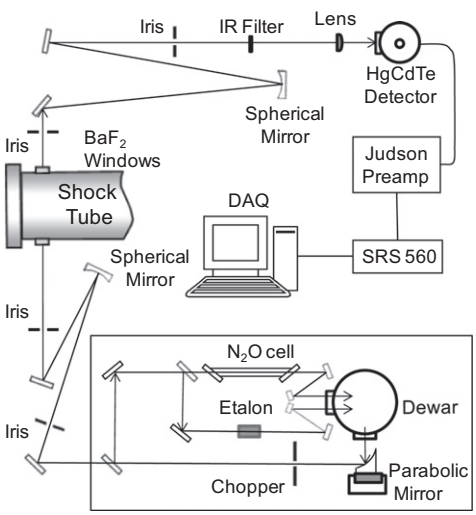

Fig. 4. Schematic of H_2O_2 laser absorption diagnostic.

50%, and added complexity. For the present study, the advantage of higher power and improved signal-to-noise ratio was considered to outweigh the advantage of greater spectral purity. Spectral scans show that the power in the side modes is roughly 10% of the primary mode and can henceforth be factored out. To further maximize power at the detector, BaF_2 windows and only gold or aluminum reflectors were used in the optical path. A long path length was established between shock-tube exit window and detector, with several stopped-down irises in between, to minimize detected emission from the shock-tube experiment.

Laser frequency calibration was typically performed using reference gas absorption lines and the transmission peaks of a germanium etalon. The free spectral range of the etalon was about 0.0475 cm^{-1} in the wavelength region of this study; however, the peak separation was not constant since lasing frequency depends nonlinearly on diode injection current. Nitrous oxide reference gas lines were always monitored in frequency calibration work, supplemented by H_2O_2 as required with occasional use of H_2O and/or NH_3 as well. A typical spectral scan is shown in Fig. 5, obtained with a temperature setting of 95.81 K. Traces are included for N_2O and H_2O_2 as well as for the germanium etalon. Analysis of the data allowed scan frequency calibration and linearization, and it is the linearized version that is shown in the figure. In this case, a H_2O_2–H_2O–argon mixture, obtained from bubbling the argon through a 30% H_2O_2/70% H_2O solution (see below), was flowing through the end wall of the shock tube under steady state conditions, and the laser beam was passed through the same shock-tube BaF_2 windows utilized for data gath-

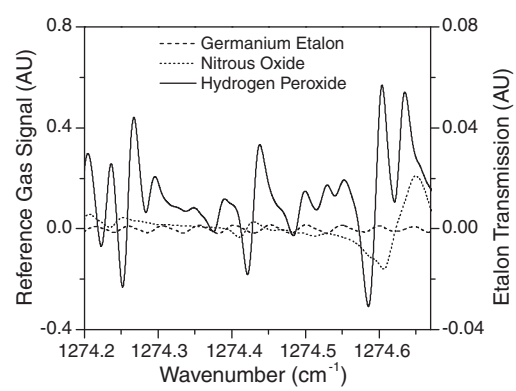

Fig. 5. Typical spectral scan of frequency space for reference NO_2 cell comparison.

ering during the actual shock experiments. Hydrogen peroxide absorption was approximately 3%, and to obtain good signal-to-noise ratio for frequency calibration scans, the laser source was frequency modulated at 2 kHz. The detected signal was subsequently demodulated and filtered using an SRS 830 lock-in amplifier. The target H_2O_2 frequency for the present study was 1274.595 cm^{-1}. This frequency corresponds to the strongest transition in the region, and is contained within the scan shown in Fig. 5.

Figure 5 shows a first-derivative spectrum, obtained by frequency modulating the laser source at 2 kHz and employing lock-in amplifiers for phase detection of the absorption signals from nitrous oxide and hydrogen peroxide. The etalon signal is not detected using a lock-in, rather it is the direct output from a HgCdTe detector that is amplified and low-pass filtered. Some noise is added to the etalon signal as a result of laser source modulation. The spectrum shown was used for frequency calibration of the laser system, and served as a vital aid in obtaining the target frequency.

A custom-made LabView program controls the laser scan and acquires data. The computer communicates with a Measurement Computing 1616HS data acquisition module through a USB interface. The program includes features to aid the frequency calibration of the obtained spectrum. Once the spectrum has been assigned, the laser is manually positioned, using the injection current, to obtain the desired frequency. Due to the narrow room-temperature linewidth of H_2O_2, combined with frequency jitter and drift of the laser system, the variation of signal amplitude while parked on a transition is substantial. Under the relatively high temperature and pressure of typical shock-tube test conditions, the broadened lineshape makes this variation insignificant.

The absolute absorption level of hydrogen peroxide must be measured before the shock experiment. To do this, the diode laser frequency is adjusted to a position slightly separated from the desired transition, and the frequency modulation (FM) amplitude is set large enough to sweep completely through the profile. The periodic output waveform, taken from the SRS560 amplifier, is used to establish the absolute amplitude of the absorption signal after correction for the baseline FM signal variation in the absence of H_2O_2. This result is divided by the amplitude of the chopped laser source to obtain an absolute peak-absorption figure for comparison with shock-tube test results.

2.3. Water laser diagnostic

A Toptica Photonics DL100L TDL configured around 1.38 μm with a DC 110 laser controller was used to measure water in the selected 1383.887-nm transition of the $\nu_1 + \nu_3$ band. This particular line was chosen for its lack of interference from neighboring transitions as outlined by Arroyo and Hanson [15] and then later confirmed at the author's facility [16]. The schematic for the water laser diagnostic is shown in Fig. 3.

Emitted light from the diode laser is first branched off a 90/10 transmitted/reflected beamsplitter, where the lesser of the two beams is sent to a wavemeter for continuous measurement. The laser wavelength was monitored by a Burleigh WA-1500 wavemeter with an uncertainty in wavelength of ±0.0001 nm. Laser frequency was controlled by diffraction grating, temperature, and current and was kept at the center of the absorption feature at 1383.887 nm throughout the experiment. Constant measurement of the baseline signal, i.e. without absorbing species, helped to eliminate inaccuracies associated with laser baseline shifts.

A long-focal-length convex lens was used to collimate the transmitted beam through sapphire windows on the shock tube. Transmitted and incident beams were made by a 50/50 beamsplitter and were both focused down into two New Focus 2317 large-area balanced germanium photoreceivers with Newport bandpass filters centered at 1384 ± 5 nm. A neutral density (ND) filter was used on the incident beam to limit the incoming signal intensity along with a linear polarizer on the transmitted beam to achieve fine adjustment of the baseline signal between experiments. Two signals were monitored, the incident intensity I_0, and the difference between the incident and transmitted intensities $I_0 - I_T$ from a SRS560 differential preamplifier. The incident beam was passed through a constantly purged, dry-N_2 environment.

The absorbance data recorded by such a setup (applicable to both the H_2O and H_2O_2 diagnostics) is defined by the Beer–Lambert relation,

$$I_T/I_0 = \exp(-k_\nu P X_{abs} L) \tag{1}$$

and describes the ratio of transmitted intensity, I_T, and incident intensity, I_0, to the spectral absorption coefficient k_ν ($atm^{-1}\,cm^{-1}$), pressure P (atm), mole fraction of the absorbing species X_{abs}, and the path length L (cm). The absorption coefficient k_ν is defined as the line strength $S(T)$ ($cm^{-2}\,atm^{-1}$), obtainable from HITRAN 2004 [11], multiplied with the lineshape function Φ_ν (cm). The lineshape function is approximated with the Voigt profile. Values for the water collisional broadening coefficient, γ_L ($cm^{-1}\,atm^{-1}$), are available for air in HITRAN [11] but not for argon. Values of the argon broadening half-width were estimated by using an air to argon γ_L ratio of 0.5, similar to results found by Negali et al. [13] near 1405 nm and Bandyopadhyay et al. [17]. Water concentration is determined before each

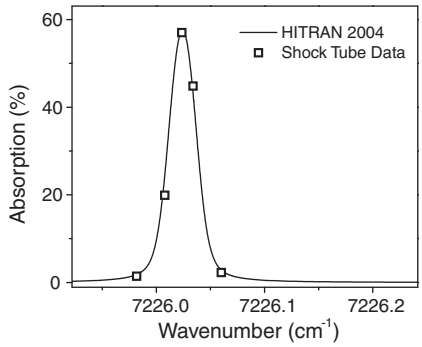

Fig. 6. Data comparison from shock tube to HITRAN 2004 predictions of water at 293 K, 5.3 torr.

experiment to calculate the hydrogen peroxide mole fraction just prior to shock arrival, a process explained in the next section.

Figure 6 shows a data comparison with the water absorption at 1383.887 nm at 293 K as determined from HITRAN 2004. Water vapor was put into the shock tube by evaporation and measured at specific frequencies around the line of interest to validate the diagnostic, as shown in Fig. 6. Excellent agreement is seen with the data determined from HITRAN and as should be expected at room temperature. This feature is well known and the data presented in Fig. 6 are meant to validate our experiment setup, therefore further sampling is not needed.

2.4. H_2O_2 delivery and vapor pressure calculation

As previously mentioned, the hydrogen peroxide present in the shock tube was carried by argon which was passed through a saturator containing a solution of 50% or 35% wt. H_2O_2, with the remainder being water. The saturator is comprised of a distilling flask made of Pyrex glassware with one inlet and one exit. Argon was passed through the solution by way of a submerged, perforated nylon 11 tube coming from the inlet of the saturator.

The saturated H_2O_2–H_2O–Ar mixture was then passed through the shock tube and into a roughing pump located at the end wall. Metering valves were put in place to control the flow rate, and the mixture was sent through the shock tube for a minimum of 30 min before running an experiment. Hydrogen peroxide decomposition is always a concern, so precaution was taken when preparing the shock tube for experiments. The present method of flowing the solution through the entire shock-tube driven section ensured that the H_2O_2 was uniform throughout the tube, and this technique proved insensitive to any potential wall reactions. Immediately prior to a shock experiment, the valves were closed and the diaphragm was broken within a minute or two (to allow the test gas to be motionless but not allowing enough time for hydrogen peroxide decomposition on the walls).

A method based on thermodynamics and Raoult's Law was used to determine the initial amount of hydrogen peroxide in the shock tube when the water concentration is known in a mixture with argon as a bath gas. The following method considers the saturated vapor pressure of hydrogen peroxide and water mixtures and takes into account the bond formed between H_2O_2 and H_2O molecules. The thermodynamics of this method were first reported by Scatchard et al. [18]. The vapor pressure of a given aqueous mixture of hydrogen peroxide and water is given as,

$$P_{v,mix} = \gamma_{H_2O} X'_{H_2O} P_{v,H_2O} + \gamma_{H_2O2} X'_{H_2O_2} P_{v,H_2O_2} \quad (2)$$

where $P_{v,mix}$ is the total mixture vapor pressure in terms of the activity coefficient γ as given by Scatchard et al. [18] and summarized by Schumb et al. [19], the known initial solution mole fraction X', and the vapor pressure of each component P_v. Relations for the vapor pressures of H_2O and H_2O_2 as well as the activity coefficients γ are provided in Supplementary data. The only two components considered are hydrogen peroxide and water, so

$$X'_{H_2O} + X'_{H_2O_2} = 1 \quad (3)$$

Knowing that this relationship holds true in the vapor pressure of the mixture, if the mole fraction of saturated water in the vapor phase with argon is known, then the relationship in Eq. (3) can be related to the mole fraction of resulting saturated hydrogen peroxide after some algebraic manipulation by

$$X_{H_2O_2,sat} = X_{H_2O,sat} P_{v,mix} / P_{v,H_2O} - X_{H_2O,sat} \quad (4)$$

For Eq. (4), $X_{H_2O_2,sat}$ represents the mole fraction of hydrogen peroxide in the vapor phase, and $X_{H_2O,sat}$ is the mole fraction of water in the vapor phase. The values for $X_{H_2O,sat}$ are known for each experiment by the diagnostic detailed in the previous section, and Eq. (4) holds valid up to a relative humidity value of 100%. For the experiments conducted in this study, the relative humidity never exceeded 50%, so this method provides an accurate analytical way of determining H_2O_2 concentration.

3. Results

Starting with the known concentration of hydrogen peroxide in the shock tube, the absorption of the laser at 7845.63 nm could be measured both before and after the passage of the incident shock wave. The intent of the present experiments was mainly to obtain a calibration of the absorption coefficient for a known level of H_2O_2 rather

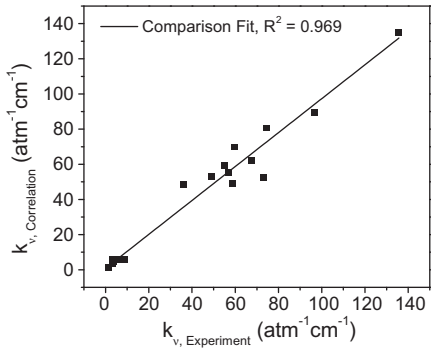

Fig. 7. Correlation comparison with experiment data.

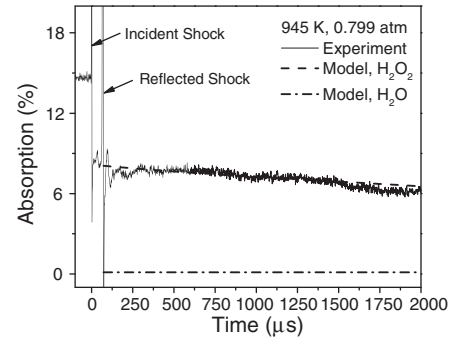

Fig. 8. Model predictions compared with shock-tube data for 2569 ppm H_2O_2. The water and hydrogen peroxide model values are as predicted from the mechanism of Davis et al. [20].

than to focus on the kinetics of the subsequent H_2O_2 decomposition behind the reflected-shock wave. Several shock-tube experiments were performed over a range of initial pressures from 0.040 to 0.071 atm (at 293 K) with $X_{H_2O_2}$ varying from 459 to 3600 ppm. Temperatures and pressures behind the incident wave varied from 683 to 1100 K and 0.41 to 0.46 atm, respectively. At the corresponding reflected-shock conditions, the hydrogen peroxide decomposed relatively quickly, so post reflected shocks were not used to generate calibration data for the hydrogen peroxide absorption coefficient. The absorption coefficient, as shown in Eq. (1), was determined from the calculated hydrogen peroxide concentrations and the transmittance recorded from the lead–salt diode laser diagnostic. These values were used to generate a correlation for k_v in the following form

$$k_v = 2.52 \times 10^4 \cdot P^{-0.963} T^{-1.02} X_{H_2O_2}^{1.145} X_{H_2O}^{-1.491} \quad (5)$$

where P and T are pressure and temperature in atm and K, respectively, with k_v in $\text{atm}^{-1}\,\text{cm}^{-1}$, and $X_{H_2O_2}, X_{H_2O}$ being mole fractions of hydrogen peroxide and water with argon in the overall mixture prior to shock. Eq. (5) offers the first calibration of its kind for use with the conditions mentioned above.

Figure 7 compares the experiment data recorded and the values obtained by the correlation at similar conditions. This correlation has a goodness of fit $R^2 = 0.969$ and assumes the primary collision partner is mostly argon. When applied to the two conditions in Fig. 2, the absorption coefficient for the 293-K, 0.04-atm example would be 83 $\text{atm}^{-1}\,\text{cm}^{-1}$, while for the 1000-K, 1-atm case k_v would be 1.06 $\text{atm}^{-1}\,\text{cm}^{-1}$.

4. Discussion and summary

Using the H_2O_2 diagnostic outlined above, the time history of hydrogen peroxide in shock-tube experiments can be monitored. For example, Fig. 8 shows 2569-ppm hydrogen peroxide decomposition with $X_{H_2O} = 0.088$ in argon at 0.79 atm behind the reflected-shock wave. In contrast to the experiment H_2O_2 signal is the low absorption by water, as predicted by HITRAN at the frequency of choice for H_2O using the predicted time history of H_2O formation from the Davis et al. [20] mechanism. The predicted absorption of water never exceeds 0.12% and thus has little interference at these conditions. The decomposition of H_2O_2 is currently predicted by the chemical kinetics mechanism of Davis et al. [20] to be in good agreement with the experiment. Note on the shock-tube data in Fig. 8, the Schlieren effect from shock wave passage is evident, which distorts the first 5 μs due to the finite length of the detector and diameter of the laser beam. Future work will incorporate improvements on the signal-to-noise ratio, where the current resolution on H_2O_2 is about 130 ppm with an absorption detection limit of 0.5%. Also, simultaneous measurements of water and H_2O_2 can be made during the entire shock wave experiment if the two different lasers are positioned at the same window location close to the end wall.

Measurements of H_2O vapor in argon without the presence of hydrogen peroxide were also performed to better characterize the effect of water absorption throughout the experiment. It was observed that there exists no interference of water absorption at initial and incident shock conditions. However, there were considerable levels of water absorption behind the reflected shock which were higher than the levels predicted by HITRAN. It was for this reason that the experiments obtained for calibration of this diagnostic included cases where the absorption from water was no more than 10% of the total absorption signal of hydrogen peroxide.

Measurements of initial hydrogen peroxide concentration and its resultant decomposition were performed in a shock tube at temperatures up to 1296 K. These measurements were con-

ducted using two laser absorption spectroscopy diagnostic systems, one at a frequency of 7226.024 cm^{-1} to measure water concentration before data recording and the other at 1274.595 cm^{-1} to measure hydrogen peroxide concentration throughout the experiment. Calibration data for mixtures of H_2O_2 ranging between 459 and 3600 ppm with varying levels of water in argon were generated by supplying a known amount of hydrogen peroxide into the shock tube. The technique of delivering specific quantities in the shock tube relies on the thermodynamics of the evaporation of water and hydrogen peroxide aqueous solutions, which results in an approximate uncertainty in initial H_2O_2 concentration of ± 65 ppm. This work resulted in a determination of H_2O_2 absorption coefficients at 1274.595 cm^{-1} at temperatures and pressures ranging from 293 K to 1100 K and 0.040 atm to 0.46 atm, respectively.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.063.

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